NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-22885/111(EC)
14 August 1992

PERFORMANCE SPECIFICATION SHEET

SWITCHES, PUSHBUTTON, LIGHT EMITTING DIODE (LED) ILLUMINATED, STANDARD, NIGHT VISION IMAGING SYSTEM (NVIS) AND SUNLIGHT READABLE DISPLAYS, SPDT AND DPDT, LOW LEVEL TO 7 AMPERES, SEALED AND UNSEALED, SOLDER AND COMMON TERMINATION SYSTEM (CTS) TERMINATIONS

This specification is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-22885.

Inches	mm
.03	0.8
.14	3.6
.20	5.1
.24	6.1
.35	8.9
.39	9.9
.52	13.2
.75	19.1
1.21	30.7

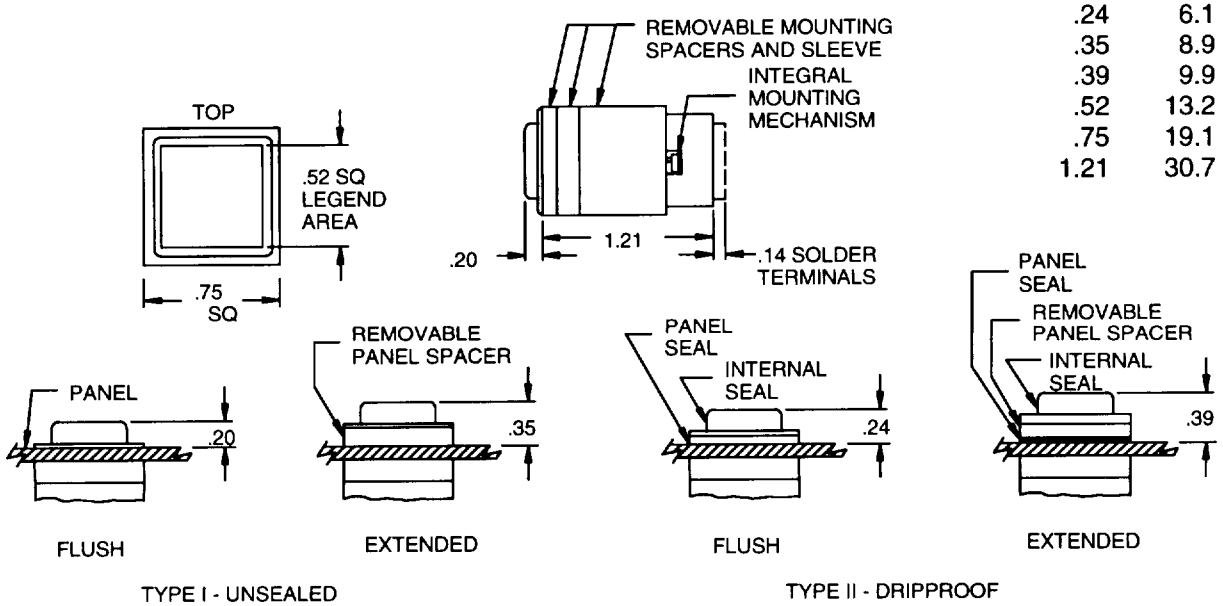
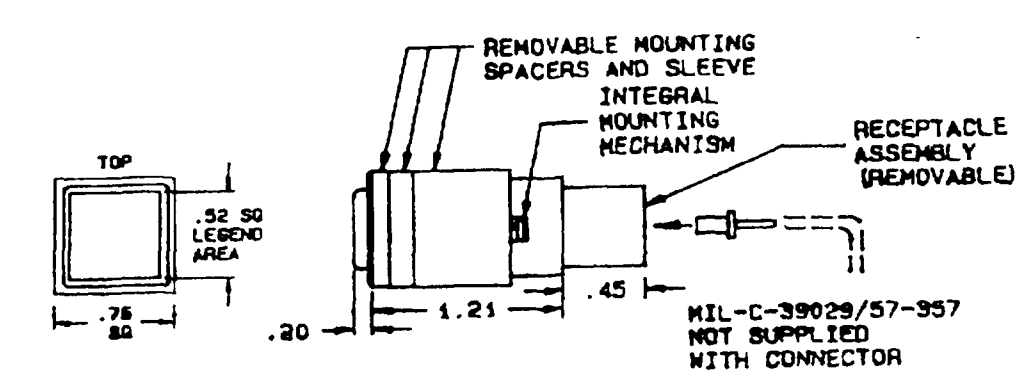


FIGURE 1. Switches - type I (unsealed, solder terminals) and type II (dripproof seal, solder terminals).

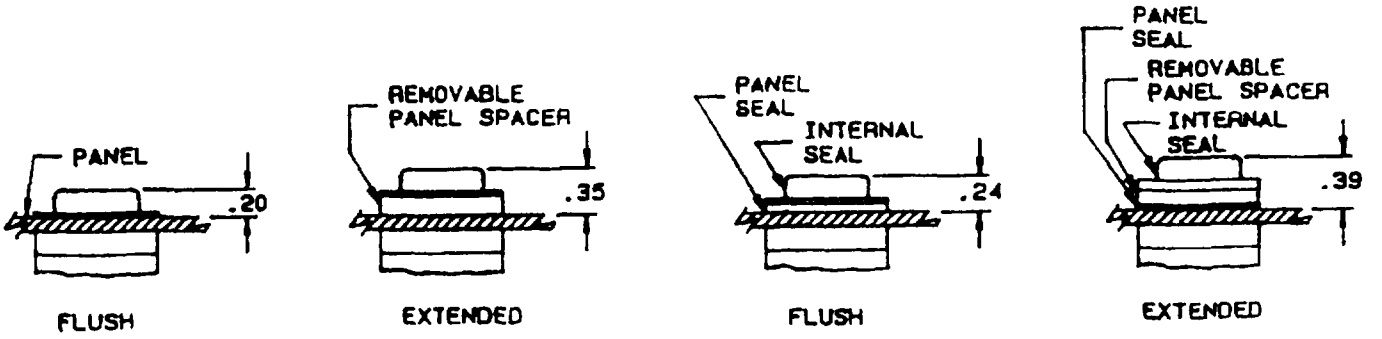
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Inches	mm
.03	0.8
.20	5.1
.24	6 1
.35	8 9
.39	9 9
45	11 4
.52	13.2
75	19 1
1.59	40 4



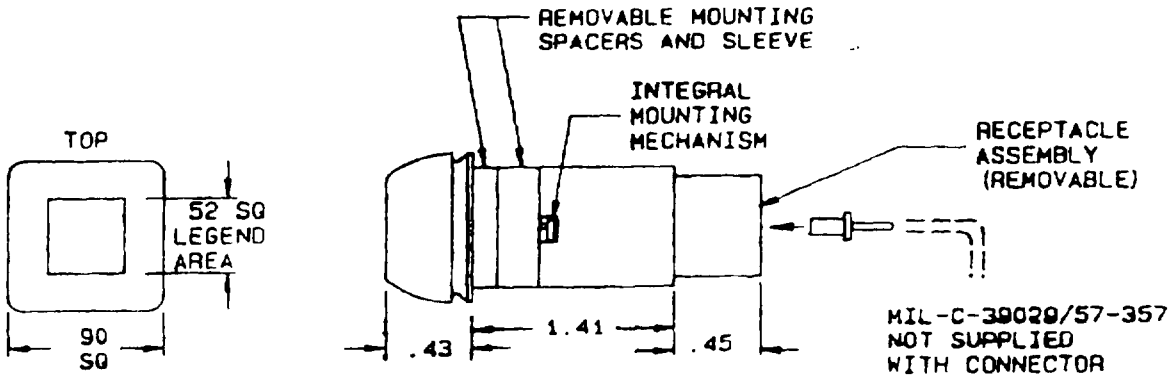
TYPE IV - UNSEALED

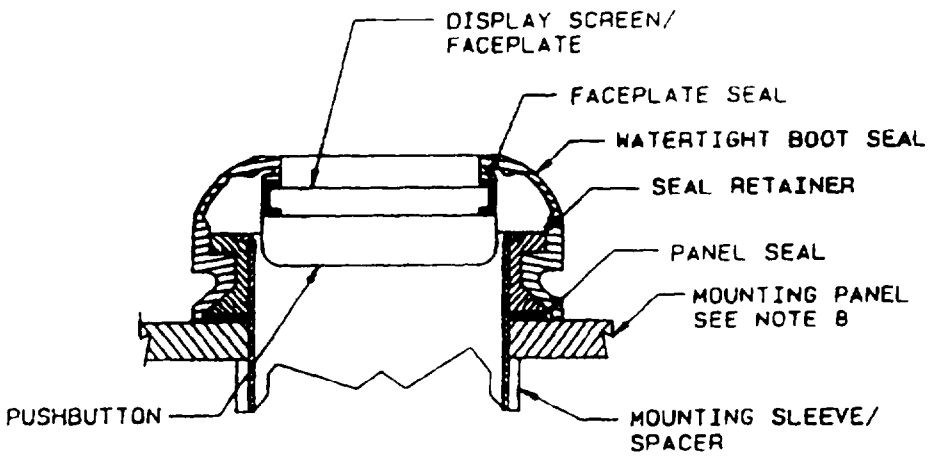
TYPE V - DRIPPROOF

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 03.
4. Pushbutton design shall provide a means to prevent incorrect installation and a means of retention with switch body during service.
5. The CTS receptacle assembly shall be removable from switch module and switch housing unit assembly
6. The switch module shall be removable from the switch housing and CTS unit assembly.
7. The CTS pins shall be M39029/57-357 and shall be removable from the CTS module assembly. The CTS pins accept 20, 22, or 24 gauge wire
8. See figure 8 for recommended mounting panel cutouts and thickness ranges.

FIGURE 3. Switches, common termination system - type IV (unsealed) and type V (dripproof seal)

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Inches	mm
03	0 8
43	10.9
.45	11 4
52	13.2
.90	22.9
1.41	35 8

SEAL CROSS SECTIONAL VIEW

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.03
4. Pushbutton design shall provide a means to prevent incorrect installation and a means of retention with switch body during service.
5. The CTS receptacle assembly shall be removable from switch module and switch housing unit assembly.
6. The switch module shall be removable from the switch housing and CTS unit assembly.
7. The CTS pins shall be M39029/57-357 and shall be removable from the CTS module assembly The CTS pins accept 20, 22, or 24 gauge wire.
8. See figure 8 for recommended mounting panel cutouts and thickness ranges.

FIGURE 4. Switch - type VI (watertight and splashproof seals, common termination system)

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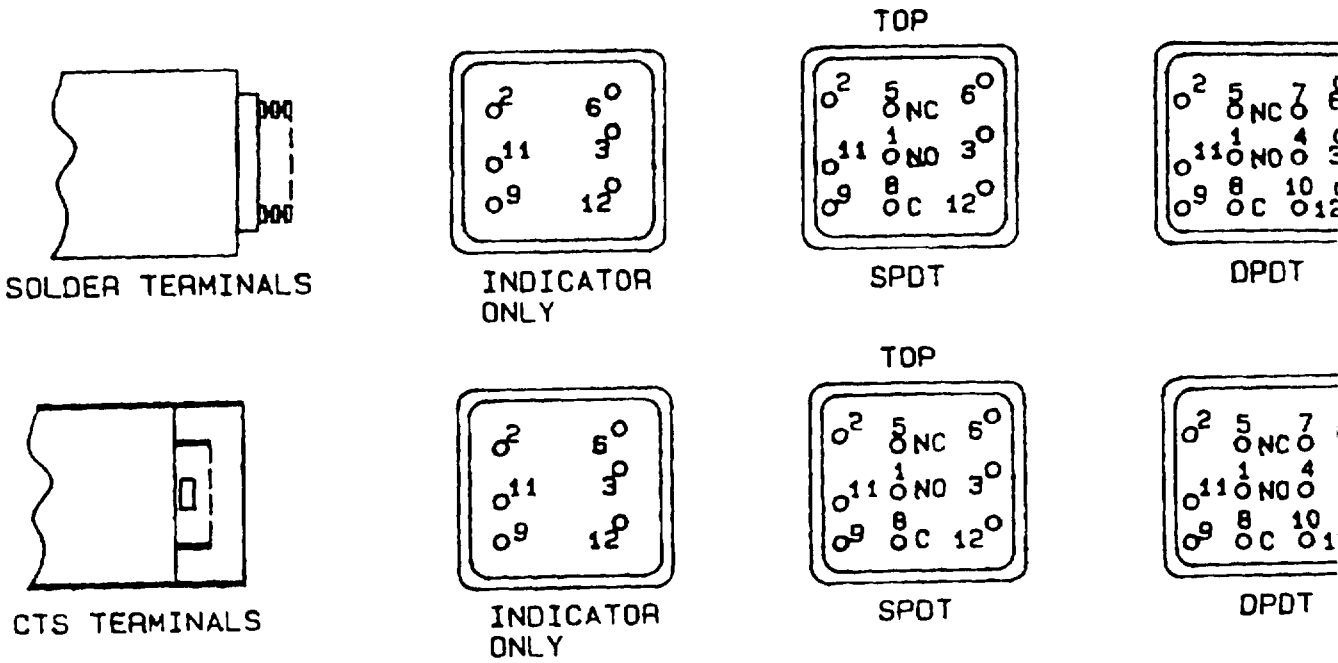
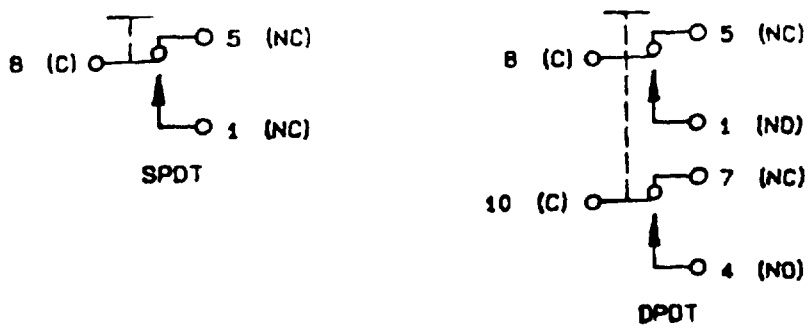


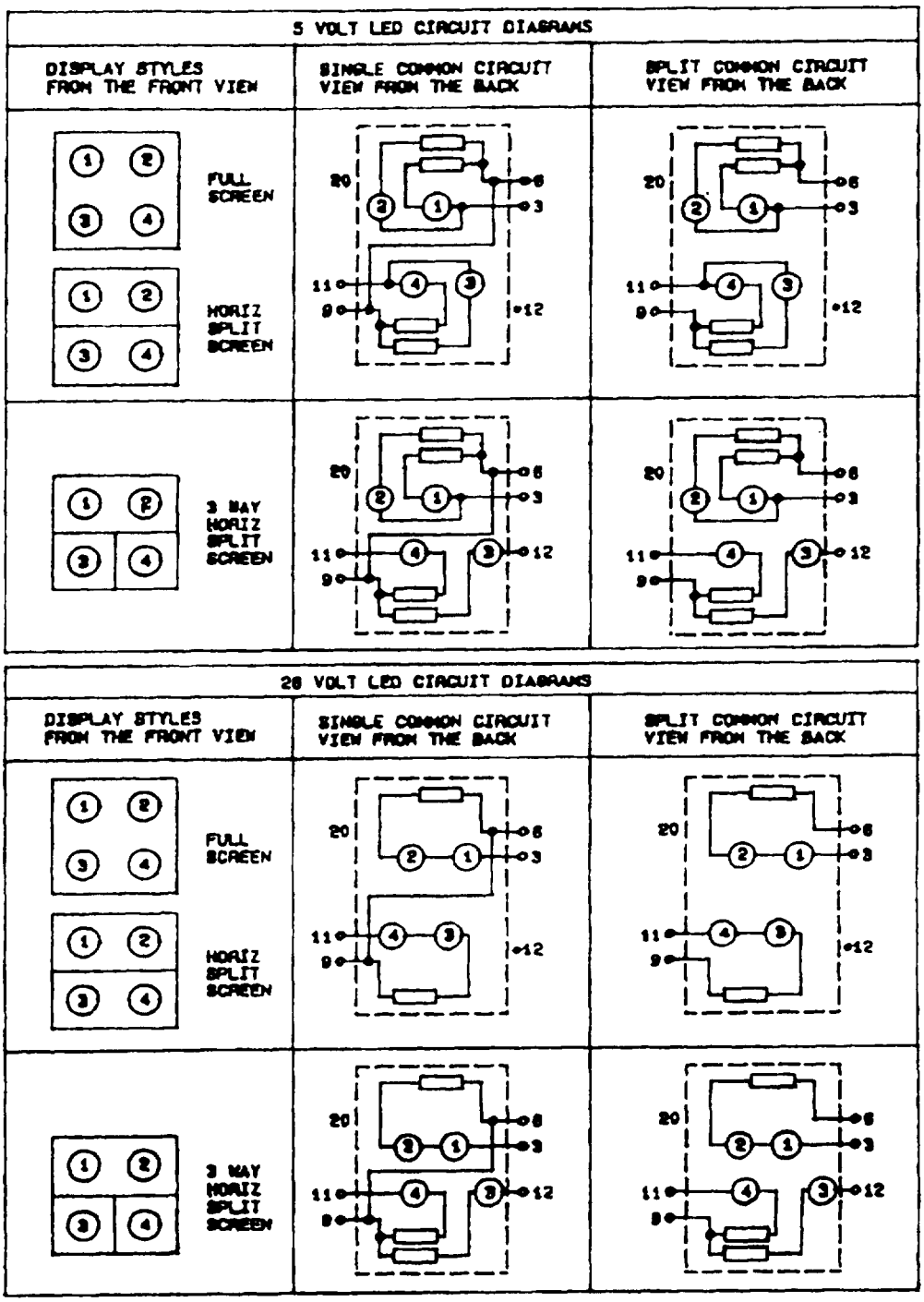
FIGURE 5 Terminal location and identification



NOTES:
1. Terminals five and seven are normally closed (NC).
2. Terminals one and four are normally open (NO).
3. Terminals eight and ten are common (C).
4. For indicator only, terminals one, four, five, seven, eight, and ten are omitted
5. For SPDT switch, terminals four, seven, and ten are omitted.

FIGURE 6 Switch circuits.

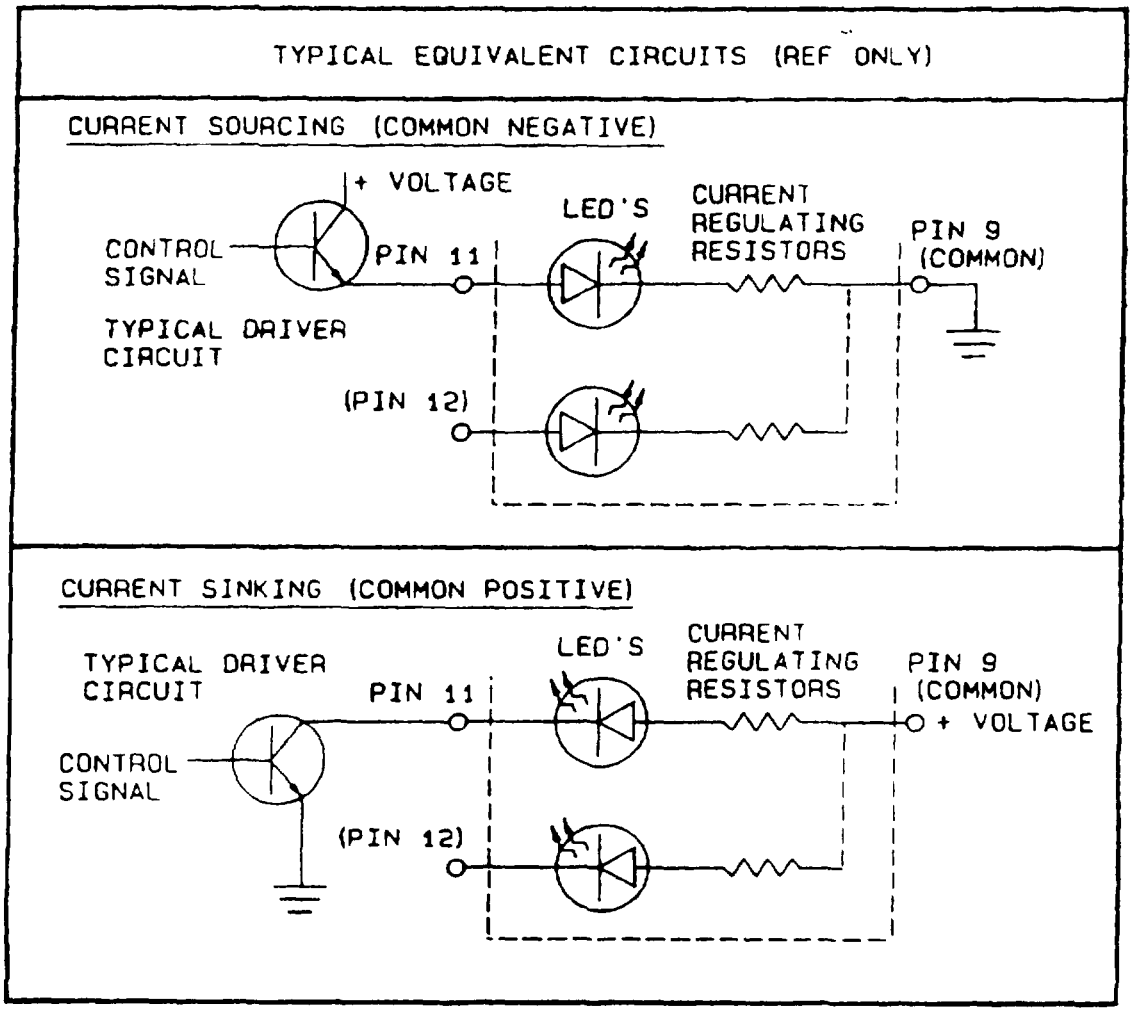
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NOTES.
1. ○ - Multi-chip LED's with position number.
2. -▭- - Current regulating resistor.

FIGURE 7 Light emiting diode circuit schematics

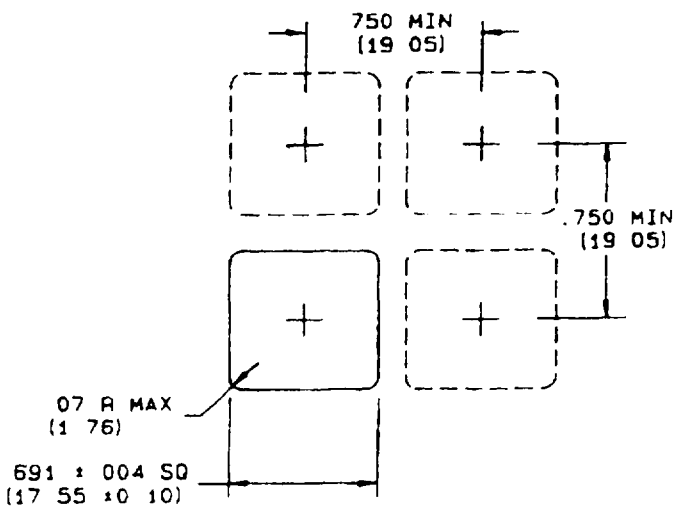
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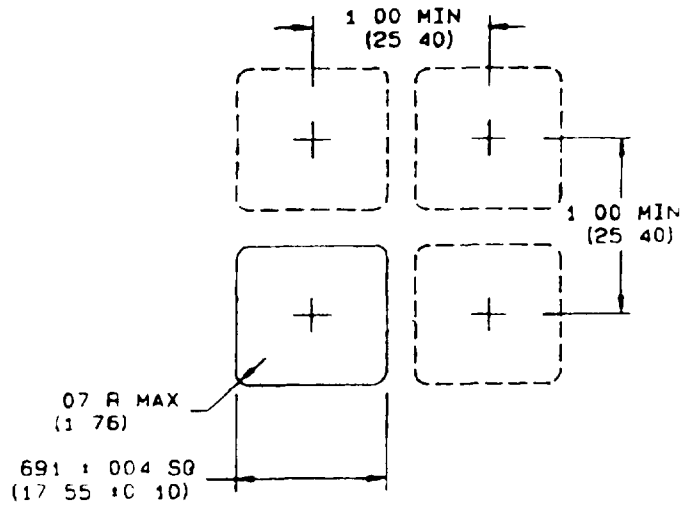
NOTE: All 5 V and 28 V LED circuits are available in either the current sourcing or current sinking configuration.

FIGURE 7. Light emitting diode circuit schematics - Continued.

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Switch body types I, II, IV, and V



Switch body types III and VI

Mounting panel thickness	
Switch body types	Panel thickness ranges
I and IV flush	.00 to .92 (0 0 to 23.4)
I and IV extended	.00 to .77 (0.0 to 19.6)
II and V flush	.00 to .88 (0.0 to 22.4)
II and V extended	.00 to .73 (0.0 to 18.5)
III and VI	.00 to .77 (0 0 to 19 6)

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only

FIGURE 8 Recommended mounting panel cutouts and thickness ranges

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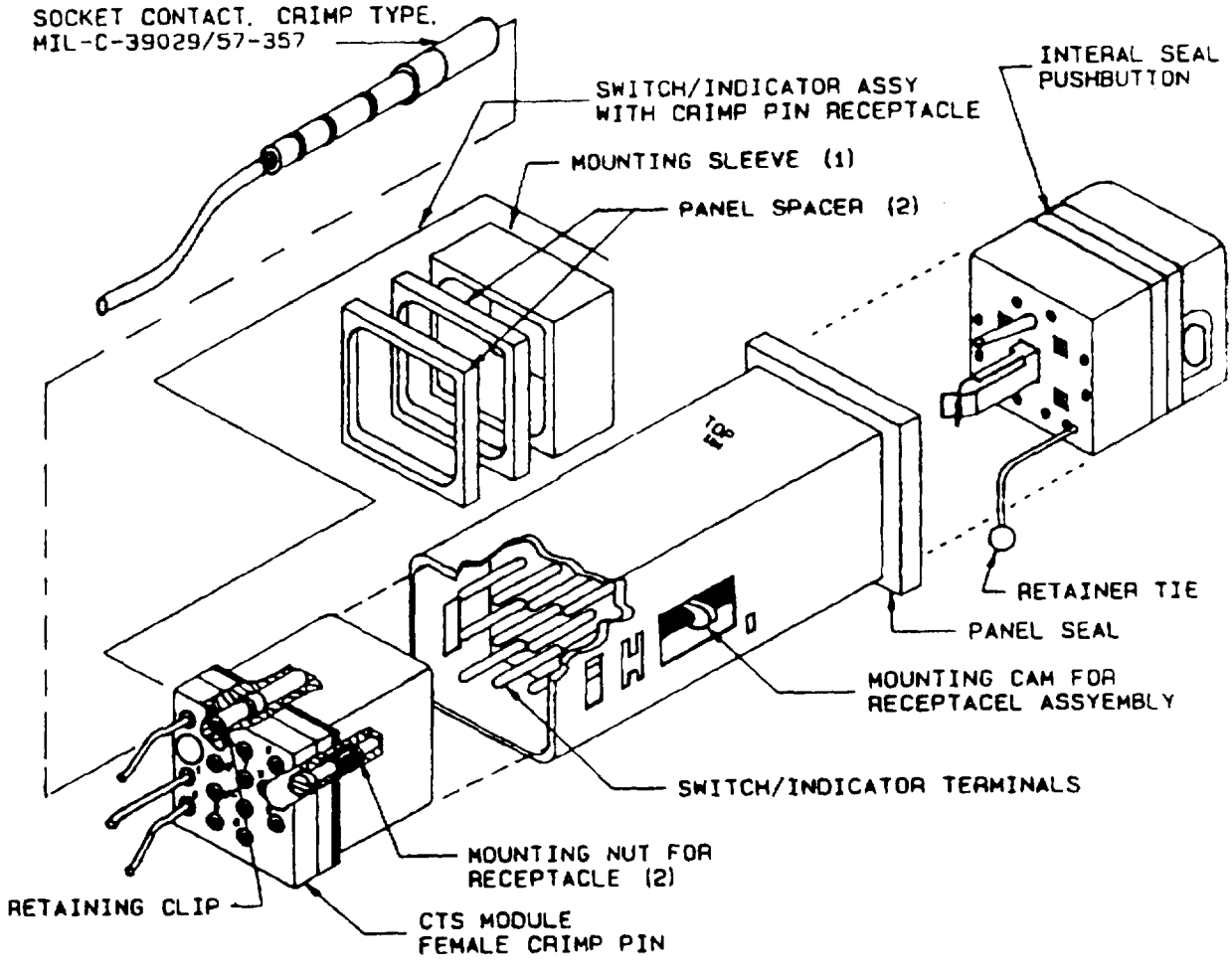


FIGURE 9 Common termination system mechanical details

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REQUIREMENTS:

Design and construction See figures 1 through 9

Complete switch and pushbutton assembly includes the following:

Switch body:

Type I switch bodies include a mounting sleeve and two panel mounting spacers See figure 1

Type II switch bodies include a mounting sleeve, two panel mounting spacers, and dripproof panel seal. See figure 1.

Type III switch bodies include a mounting sleeve, panel mounting spacer, splashproof/watertight seal, seal retainer, and gasket See figure 2

Type IV switch bodies include a mounting sleeve, two panel mounting spacers, and CTS module/receptacle. See figure 3.

Type V switch bodies include a mounting sleeve, two panel mounting spacers, dripproof panel seal, and CTS module/receptacle See figure 3

Type VI switch bodies include a mounting sleeve, panel mounting spacer, splashproof/watertight seal, seal retainer, gasket, and CTS module/receptacle See figure 4

Pushbutton All pushbutton assemblies include a display faceplate and legend assembly, display segment divider, nonreplaceable LED lamps, lamp contact board assembly, pushbutton faceplate seal, retainer/bezel, and captive pushbutton retainer

Other parts, when applicable, are NVIS compatible lighting, EMI/RFI shielding, and dripproof or splashproof/watertight seals.

Enclosure designs:

Symbol	Switch type
1 (unsealed)	I and IV
2 (dripproof)	II and V
3 (watertight) and 4 (splashproof)	III and VI

Temperature characteristic Symbol 1 (-55°C to +85°C)

Vibration grade Symbol 3 (10 to 2,000 Hz, 15 g peak)

Operation

Symbol A (momentary action)
Symbol B (alternate action).
Symbol H (indicator light)

Display types

N (Legend is white on nonreflective matte black background when LED light is not lighted Legend is colored on black background when LED is lighted)

W (Legend is opaque black on white background when LED lamp is not lighted Legend is opaque black on colored background when LED is lighted)

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S (Sunlight readable contrast requirements apply Background is nonreflective matte black Legend is not visible when LED is not lighted Legend appears in color on black background when LED is lighted)

H (Hidden legend. Same as display type S except contrast requirements do not apply)

Material

Finish Aluminum alloy may be plated with black chrome over nickel in accordance with MIL-C-14538

Operating characteristics:

Actuating force:

Unsealed 2 pounds ±1 pound

Dripproof. 3 pounds ±1 pound

Splashproof/watertight: 4 pounds ±1 pound.

Actuating travel: 125 inch maximum, 0.08 inch minimum

Pushbutton removal force 2 to 4 pounds

Shock

Switch types I, II, IV, and V. Method I

Switch types III and VI Method I and method II (high-impact) applicable For momentary action switches, during the high-impact shock test, there shall be no opening of closed contacts or closing of open contacts in excess of 40 ms. For alternate action switches, contacts may transfer during high-impact shock test, and switch contact stability need not be monitored during the test.

Salt spray: In accordance with MIL-S-22885, except the length of test shall be 48 hours (MIL-STD-202, method 101, test condition B)

Sand and dust. Applicable

Electrical endurance 25,000 cycles at the following electrical ratings

Silver contacts.	Sea level	50,000 feet
28 V dc:		
Resistive	7.0 A	4.0 A
Inductive	4.0 A	2.5 A
Lamp	2.5 A	2.5 A
115 V ac, 60 Hz:		
Resistive	7.0 A	
Inductive	7.0 A	
Lamp	2.0 A	
Gold plated contacts.		
28 V dc:		
Resistive	1.0 A	
Inductive	0.5 A	

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Low-level life Applicable to switches with gold plated contacts

Intermediate current Applicable to switches with silver contacts

Mechanical endurance 100,000 cycles consisting of 5,000 cycles of operation at -55°C ±2°C, 10,000 cycles of operation at +85°C ±2°C, and 85,000 cycles of operation at room temperature

Illuminated color and luminance See table I

TABLE I. Illuminated color and luminance limits

Color	Color code	Color limits 1/		Minimum average luminance 2/
		x	y	
Red	RD	.680 .687 719 .712	306 .313 .281 274	70 foot-lamberts
Amber yellow	AY	512 519 592 .585	473 480 408 401	85 foot-lamberts
Lime green	LG	416 423 .492 485	568 .575 .507 .500	170 foot-lamberts
Green	GR	.295 302 .395 388	685 692 603 596	55 foot-lamberts
Orange	OR	.620 627 666 .659	366 373 334 .327	70 foot-lamberts
Ultra red	UR	680 .687 .719 712	306 .313 .281 .274	200 foot-lamberts
NVIS green B	JJ	In accordance with MIL-L-85762		0.1 foot-lamberts at 0.2 percent duty cycle 3/
NVIS yellow	KK			15 foot-lamberts at 50 percent duty cycle 3/
NVIS red	LL			15 foot-lamberts at 100 percent duty cycle 3/

1/ Corners of the x and y envelope limits from the CIE chromaticity diagram
2/ With and without EMI/RFI shielding.
3/ Appropriate power modulation required to regulate LED illumination output

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EMI/RFI shielding When applicable, the shielding efficiency test of MIL-S-22885 shall be conducted at the frequencies and with the minimum attenuation requirements as specified below

Electric field	Minimum attenuation	Plane wave	Minimum attenuation
150 kHz	60 dB	400 MHz	39 dB
500 kHz	60 dB	1 GHz	33 dB
1 MHz	60 dB	2.5 GHz	28 dB
10 MHz	60 dB	10 GHz	20 dB

Sunlight readability Applicable to display type S switches Units shall be tested for θ_1 = 45°C and θ_2 = 0°C only This procedure does not test readability in direct reflected specular sunlight (glare) The average contrast ratio of each lighted legend character to the background shall be as specified in table II In the unlighted legend condition, the average contrast ratio of each legend character to the background shall have an absolute value ≤ 0.1 The legend used for testing shall have a character height of approximately 0.09 to 0 1 inch. The stroke width shall be characteristic of each manufacturer's standard military product

TABLE II Contrast ratio requirements for sunlight readable displays

Color code	Legend/LED color	Contrast ratio 1/	
		Minimum lighted	Maximum unlighted
UR	Ultra red	90	1
LG	Lime green	.60	1

1/ With and without EMI/RFI Shielding

NVIS compatibility When switches are tested as specified below, the chromaticity, luminance and spectral radiance output of the illuminated display shall be NVIS compatible as specified in MIL-L-85762 The following equipment requirements apply:

NVIS green B Shall meet all class A and class B equipment requirements of MIL-L-85762 for illuminated controls

NVIS yellow Shall meet type I class A, type II class A, and type II class B equipment requirements of MIL-L-85762 for caution signals.

NVIS red: Shall meet type I class B, type II class A, and type II class B equipment requirements of MIL-L-85762 for warning signals.

The test procedure for measuring NVIS luminance, chromaticity, and spectral radiance shall be in accordance with MIL-L-85762 for illuminated displays (for NVIS green B), caution signals (for NVIS yellow), and warning signals (for NVIS red).

Touch temperature: Applicable to all body types of LED illuminated switches

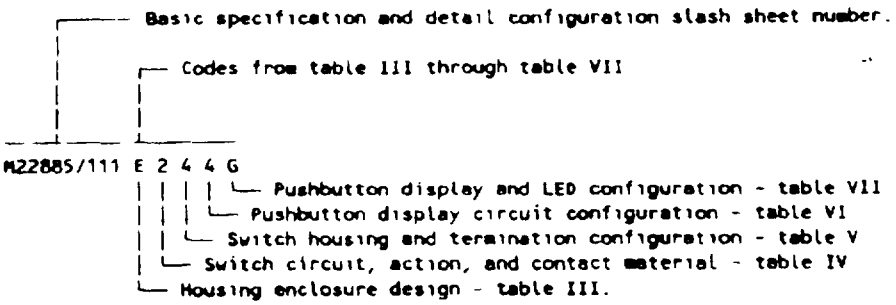
Requirements. When switches are tested as specified below, the maximum difference between the stabilized lens face temperature and the ambient temperature shall not exceed +25°C

Test method The test method shall be in accordance with EIA-448-20 using the recommended panel cutout (see figure 8) The test shall be performed with each of the standard LED voltages at full rated current and at 100% duty cycle

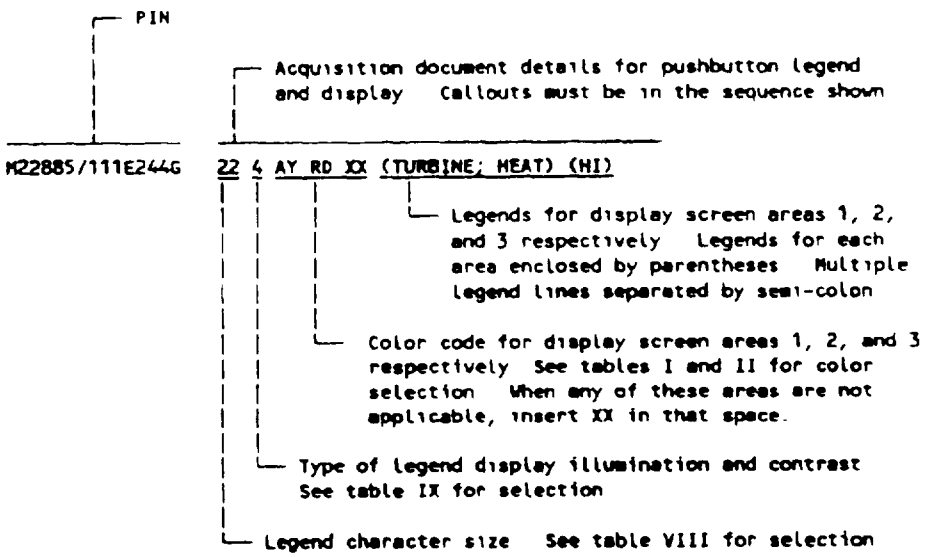
Marking All required marking shall be permanent

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Part or Identifying Number (PIN). The PIN shall be constructed as illustrated in the following example:



Legend details and display screen colors are not defined by the PIN. These details are specified by user's acquisition documents in accordance with the acquisition requirements for category II switches, see section 6 of MIL-S-22885. For the convenience of users, an extension to the PIN may be utilized on the acquisition document to define the legend details and colors. The method of constructing this extension is illustrated below. The extension is for convenience only, and is not a part of the PIN.



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TABLE III Codes for housing enclosure design

Enclosure design	Without EMI/RFI shielding		With EMI/RFI shielding	
	75 G shock	Hi-impact shock	75 G shock	Hi-impact shock
With solder terminations				
Unsealed	A	---	J	---
Dripproof	B	---	K	---
Watertight/splashproof	C	S	L	W
With CTS terminations				
Unsealed	E	---	N	---
Dripproof	F	---	O	---
Watertight/splashproof	G	U	P	Y

TABLE IV Codes for switch circuit, action, and contact material

Switch circuit	Contact material	
	Silver	Gold
Momentary action.		
Single pole, double throw (SPDT)	1	5
Double pole, double throw (DPDT)	2	6
Alternate action:		
Single pole, double throw (SPDT)	3	7
Double pole, double throw (DPDT)	4	8
Indicator	---	0

TABLE V. Codes for termination type.

Termination type	Code
Turret, solder	1
Common termination system (CTS):	
Without termination module 1/	3
With termination module 2/	4

1/ Designates CTS switch housing without connector module. Connector modules may be supplied pre-wired to wire harness assembly.

2/ Designates CTS switch housing with connector module installed. Sealing plugs and termination pins not included.

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TABLE VI Pushbutton display module LED circuit configuration

Load circuit configuration	Code	
	Single common	Split common
Current sourcing	1	3
Current sinking	2	4

TABLE VII Pushbutton display screen and LED design

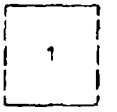
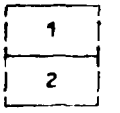
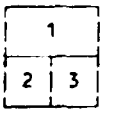
LED voltage	LED light type	Display screen design code		
		1	1 / 2	1 / 2 | 3
5 V dc	Standard	A	G	N
	Sunlight readable 1/	B	H	P
	NVIS compatible 2/	C	J	R
28 V dc	Standard	D	K	S
	Sunlight readable 1/	E	L	T
	NVIS compatible 2/	F	M	U

1/ Available in ultra red and lime green only. See application information for contrast ratio data on hidden legend displays in other colors.

2/ Available in display type H only (hidden legend), see table IX

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TABLE VIII Legend character size 1/ 2/ 3/ 4/

Size code	Character size (inches)	Display screen area	Number of characters per legend area		
			1	2	3
10	None		None	X	X
11	072		5 x 12	X	X
12	087		4 x 10	X	X
13	.100		3 x 9	X	X
14	125		3 x 7	X	X
15	.145		2 x 6	X	X
20	None		None	None	X
21	.072		2 x 12	2 x 12	X
22	.087		2 x 10	2 x 10	X
23	.100		1 x 9	1 x 9	X
24	125		1 x 7	1 x 7	X
25	145		1 x 6	1 x 6	X
30	None		None	None	None
31	.072		5 x 12	2 x 6	2 x 6
32	087		4 x 10	2 x 5	2 x 5
33	100		3 x 9	1 x 4	1 x 4
34	125		3 x 7	1 x 3	1 x 3
35	145		2 x 6	1 x 2	1 x 2

1/ Alpha-numeric character format is alternate gothic number 2, Sans-serif
2/ Number of characters shown above is number of lines times number of characters per line For example, 5 x 12 is 5 lines of 12 characters each
3/ Maximum number of characters per line shown above is an approximate value The actual number of characters may be more or less, depending on characters used
4/ Size codes 10, 20, and 30 are not applicable to legend codes 6 and 7 of table IX

TABLE IX. Legend display and illumination details

Code	Nonilluminated		LED illuminated		MIL-S-22885 display type
	Legend	Background	Legend	Background	
4 1/	White	Black	LED color	Black	N
5 1/	Black	White	Black	Lamp Color	W
6 2/	Hidden	Black	LED color	Black	S
7 3/	Hidden	Black	NVIS or LED color 3/	Black	H

1/ Not available in NVIS compatible colors.
2/ Sunlight readable Available in colors ultra red and lime green only Not NVIS compatible.
3/ Hidden legend Available in NVIS colors or standard color codes OR, AY, RD, and GR See application information for approximate contrast ratios of standard colors Not sunlight readable

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Inspections

LED failure LED failure during any inspection shall constitute a test failure

Qualification:

Group submission: Table X applies, provided that the switching elements used are qualified MIL-S-8805 category I or II basic sensitive switches and the endurance and electrical ratings are compatible

TABLE X Qualification inspection - group submission

Test sample	Qualification inspection table of MIL-S-22885		Extent of approval
	Group	Number of samples	
M22885/111B213D	I	12	All PIN's
	II 1/	4 from group I	
	III	2 from group I	
	VI 2/	2 from group I	
	VII	4 from group I	
M22885/111A822A	I 3/	6	
	II 1/	4 from group I	
	VII 4/	2 from group I	
M22885/111Y241K	I	12	
	II 5/ 6/	4 from group I	
	III	2 from group I	
	V 6/	2 from group I	
	VI 7/	2 from group I	
	VII 6/ 8/	2 from group I	
	VIII 9/	42 10/	

1/ Shock. Method I
2/ Electrical endurance Sea level, inductive dc load only
3/ Test current and voltage for contact resistance and operating characteristics tests shall not exceed 10 mA and 30 mV dc on all gold plated contact switches subject to low-level life load tests Low-level life contact resistance requirements apply.
4/ Mechanical endurance test not required.
5/ Shock: Method II only.
6/ Sealing shall be verified by the watertight and splashproof tests only.
7/ Electrical endurance: Sea level, resistive dc load only.
8/ Intermediate current test not required.
9/ NVIS color and radiance measurements shall be made, as applicable, during group VIII inspections

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TABLE X. Qualification inspection - group submission - Continued

10/ In combination with the test sample specified in the first column, 42 pushbutton display screen assemblies shall be tested Two each of six colors for each standard display type, N and W, two each of two colors for display type S, and two each of three NVIS colors and four standard colors for display type H One of the pushbutton samples for each color/display type combination shall be EMI/RFI shielded, and the other shall be unshielded One of the pushbutton samples for each color/display type combination shall have 5 V LED's, and the other shall have 28 V LED's Test legend characters and display screen design shall be as follows

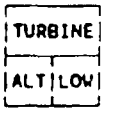


TABLE XI Group A inspection

Seal (when applicable) 1/
Visual and mechanical examination
Operating characteristics
Dielectric withstanding voltage
Contact resistance 2/
Sunlight readability (when applicable)
NVIS compatibility (when applicable)

1/ Visual in-process inspection of seal with high intensity light source may be used.

2/ Low level test current and voltage may be used for contact resistance tests on gold contact switches The test current and voltage shall not exceed 10 mA and 30 mV dc The maximum contact resistance under this condition is 3 ohms

APPLICATION INFORMATION

Storage temperature -65°C to +100°C

U.L. listing: 250 V ac, 60 Hz, 7 amperes, listing L7 (21 SKB3-H J40)

Oil, fuel, and solvent resistance: The boot seal and the faceplate assembly of switch body types III and VI (splashproof/watertight sealed) are generally resistant to hydraulic and lubricating oils, diesel oil, jet fuels, gasoline, kerosene, methanol, and most solvents including trichlorethylene, perchlorethylene, and isopropyl alcohol. (A listing of specific types of chemicals, recommended and not recommended, may be obtained from the switch manufacturer.)

Category II product offerings:

Printed circuit terminals (switches with terminals designed for direct insertion into printed circuit boards).

Contrast ratios for hidden legend displays: See table XII. Contrast ratios specified in table XII are calculated using the sunlight readability test procedure but do not meet minimum contrast ratio requirements to be considered sunlight readable

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TABLE XII. Average contrast ratios for hidden legend displays (not sunlight readable)

Color code	Legend/LED color	Contrast ratio	
		Minimum lighted	Maximum unlighted
OR	Orange	.25	1
AY	Amber yellow	20	1
RD	Red	20	1
GR	Green	.20	1

CONCLUDING MATERIAL

Custodian
Navy - EC

Review activity
DLA - ES

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5930-N686)

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